Projectors, Packets, and Transfer

Resolution, conservation, and the passage to infinite systems

Resolution, dynamics, and reconstruction

A physical description often begins by separating a state into spatial cells, frequency bands, spin channels, tensor factors, or spectral sectors. Such a separation is useful only when the components can be reconstructed, their interactions can be described, and different resolutions can be compared. Those requirements lead to the calculus developed here.

An orthogonal projector selects a packet. Placing an operator between two projectors gives a transfer block. A skew-adjoint generator determines an antisymmetric current between packet weights. Grouping the labels produces a coarser description in which internal currents cancel. These statements can be understood in a small matrix example before any infinite-dimensional analysis is needed.

The same questions become more demanding when frames move, states become mixed, or the system exchanges information with an environment. Countable resolutions introduce convergence. Unbounded operators introduce domains. Each extension retains part of the finite algebra and adds a hypothesis that must remain visible where it is used.

The free Dirac construction that follows gives these ideas a concrete analytic realization. Fourier modes resolve a spinor field, finite matrices govern its components, and completion leads to a self-adjoint Hamiltonian. Gauge transport, constrained gravity, and arithmetic refinement then test which parts of the same language survive in different settings.

> **The organizing question**
>
> Which quantities can be reconstructed exactly, which balances survive a change of resolution, and which additional hypotheses are required for an infinite or unbounded system?

The mathematical arguments use ordinary complex linear algebra, operator theory, and Hilbert-space analysis. Formal source correspondences and verification records are collected in the technical appendices. Numerical illustrations explain specified examples rather than replacing universal proofs.

Reading routes and conventions

The first route follows the algebra. Begin with orthogonal resolution and block composition, then read the current identity, coarse aggregation, and moving frames. These sections provide the common notation for the rest of the volume. The worked examples keep the states and operators explicit.

The second route follows physical descriptions. Mixed states, tensor marginals, and Lindblad jumps show which information a packet weight retains and which information still resides in coherence. The positive-effect construction then explains how an overlapping resolution can be represented inside a larger sharp one. Classical probabilities arise only after the relevant normalization and closure assumptions have been supplied.

The third route follows analysis. Countable resolutions require an actual summation statement. An unbounded operator requires its domain. A strong limit acts on each fixed vector, while an operator-norm limit controls all unit vectors at once. The final chapters make these distinctions explicit and explain why the finite pointwise algebra $\mathbb{C}^N$ cannot support a nonzero complex-linear derivation obeying the ordinary Leibniz rule, even though matrix corners support nontrivial commutators.

Convention	Meaning
Inner product	$\langle x, y\rangle$ is conjugate-linear in x and linear in y.
Projector	$P_a = P_a^* = P_a^2$, with orthogonality between distinct packet labels.
Transfer block	$L_{ba} = P_b L P_a$ takes a component in packet a to packet b.
State equation	$\dot{\psi} = K\psi$, with $K^* = -K$. The Hamiltonian convention is $K = -iH$ in units $\hbar = 1$.
Packet weight	$p_a = \|P_a\psi\|^2$. It is a probability for a normalized state and a complete orthogonal resolution. It is not an energy expectation by definition.
Current orientation	$J_{b\leftarrow a}$ is inflow into packet b from packet a, so $\dot{p}_b = \sum_a J_{b\leftarrow a}$ in the finite closed setting.
Mixed state	$\rho \geq 0$, $\operatorname{tr}\rho = 1$ in the finite-dimensional sections.
Coarse cell	A union of disjoint fine packet labels. Its projector is their sum, with the relevant convergence and domain hypotheses in the countable setting.
Unbounded action	An expression such as $H\psi$ includes the condition $\psi \in \mathcal{D}(H)$.
Time and units	K has units of inverse time. We set $\hbar = 1$ when identifying $K = -iH$. All plotted examples state their numerical parameters.

Finite-dimensional statements appear with that hypothesis. A finite projector family can also act on an infinite-dimensional Hilbert space, provided the operators and differentiability assumptions in the statement make sense. These are different uses of the word finite.

The three companions serve different purposes. The monograph gives the mathematical argument. The visual atlas makes specific states, currents, and convergence examples visible. The Lean companion records exact declarations and evidence. A displayed ordinary proof is not silently promoted to a newly compiled theorem, and a sampled curve is not used to establish a universal limit.

Finite resolutions and dynamics

1 Resolving a state into orthogonal packets

A packet begins with a choice of resolution. One may separate a field by spatial cells, a vector by coordinate groups, or a wave by spectral bands. The meaning of the labels changes between applications. The algebra below depends on the relations between the projectors and on the operators acting between their ranges. It does not require a preferred coordinate basis inside each range.

Throughout this section, $\mathcal{H}$ is a complex Hilbert space and I is a finite index set. The inner product is conjugate-linear in its first argument and linear in its second. Every operator initially belongs to $\mathcal{B}(\mathcal{H})$, the bounded operators defined on all of $\mathcal{H}$. The dimension of $\mathcal{H}$ may be infinite. Finiteness refers to the number of packets. Countable resolutions and unbounded generators require the additional hypotheses developed later.

Definition 1.1 (Finite orthogonal atlas)**.** A finite orthogonal atlas is a family $(P_a)_{a\in I}$ satisfying

$$P_a^* = P_a, \qquad P_a P_b = \delta_{ab} P_a, \qquad \sum_{a\in I} P_a = \mathrm{id}. \tag{1}$$

Its packet of a vector ψ is $\psi_a = P_a\psi$. Its packet weight is $p_a = \|\psi_a\|^2$. Zero projectors are allowed, so a label can describe an empty sector.

The word weight avoids assigning physical units before an application supplies them. If $\|\psi\| = 1$, these weights form a probability distribution. If the state carries another normalization, they sum to its squared norm. They are not automatically the energy expectation of a Hamiltonian. For example, a normalized energy eigenvector always has total packet weight one, regardless of its eigenvalue.

Proposition 1.2 (Reconstruction and norm separation)**.** *For every $\psi, \phi \in \mathcal{H}$,*

$$\psi = \sum_a \psi_a, \qquad \langle \psi, \phi\rangle = \sum_a \langle P_a\psi, P_a\phi\rangle, \qquad \|\psi\|^2 = \sum_a \|\psi_a\|^2. \tag{2}$$

Packets with different indices are orthogonal, including packets obtained from different vectors.

Proof. Evaluate $\sum_a P_a = \mathrm{id}$ on ψ. For $a \neq b$, self-adjointness and orthogonality give

$$\langle P_a\psi, P_b\phi\rangle = \langle \psi, P_aP_b\phi\rangle = 0.$$

Expand both entries of $\langle\psi,\phi\rangle$ using the finite reconstruction. All terms with unequal indices vanish. Taking $\phi = \psi$ proves the norm identity. □

This proof separates three roles. Completeness reconstructs the vector, self-adjointness relates the projections to the inner product, and pairwise orthogonality removes cross terms. A mere decomposition of the identity by arbitrary linear maps does not imply the norm formula. A family of positive effects supplies a different probability construction, discussed later, because effects need not be idempotent or pairwise orthogonal.

Example 1.3 (Two packets inside four coordinates)**.** Let $\mathcal{H} = \mathbb{C}^4$ with its usual inner product, and set

$$\psi = \frac{1}{\sqrt{7}}(1, i, 2, -i)^{\mathsf{T}}, \qquad P_A = \mathrm{diag}(1,1,0,0), \quad P_B = \mathrm{diag}(0,0,1,1).$$

Then $\psi_A = (1, i, 0, 0)^{\mathsf{T}}/\sqrt{7}$ and $\psi_B = (0, 0, 2, -i)^{\mathsf{T}}/\sqrt{7}$. The weights are $p_A = 2/7$ and $p_B = 5/7$. Refining into the four coordinate projectors gives weights $1/7, 1/7, 4/7, 1/7$. The two descriptions resolve the same vector at different levels. The coarse description still contains the two-dimensional packet vectors. Retaining only the two scalar weights would discard their internal phases and amplitudes.

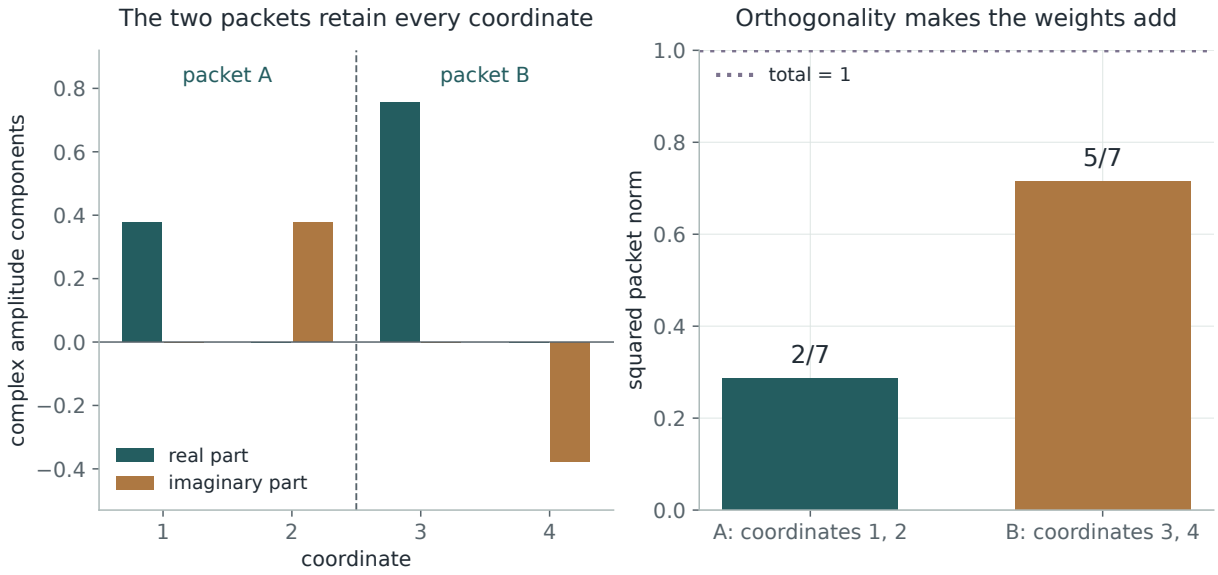


Figure 1. The four-coordinate state of Example 1.3. Packet vectors retain amplitudes and phases, while their two weights retain only the squared sizes.

2 Operator blocks and exact reconstruction

The same atlas resolves an operator on both its input and output sides. For $X \in \mathcal{B}(\mathcal{H})$, define

$$X_{ba} = P_b X P_a. \tag{3}$$

The first index names the destination and the second names the source. As a map between packet spaces, this is $X_{ba} : \operatorname{Ran} P_a \to \operatorname{Ran} P_b$. As a bounded operator on $\mathcal{H}$, it is extended by zero on the orthogonal complement of $\operatorname{Ran} P_a$. Keeping these two descriptions aligned prevents a block from being composed with an incompatible input space.

Proposition 2.1 (Block calculus)**.** *For bounded X, Y,*

$$X = \sum_{b,a} X_{ba}, \qquad P_b X \psi = \sum_a X_{ba} \psi_a, \tag{4}$$

$$(XY)_{ca} = \sum_b X_{cb} Y_{ba}, \qquad (X^*)_{ab} = (X_{ba})^*. \tag{5}$$

The identity has diagonal blocks P_a and zero off-diagonal blocks. If $X = X^*$, its opposite blocks are adjoints.

Proof. Insert $\sum_b P_b = \mathrm{id}$ to the left of X and $\sum_a P_a = \mathrm{id}$ to its right. This gives the first formula. Evaluate the right resolution on ψ for the second. For the product,

$$P_c XY P_a = P_c X\Big(\sum_b P_b\Big) Y P_a = \sum_b (P_c X P_b)(P_b Y P_a),$$

where $P_b^2 = P_b$ supplies the repeated middle projector. Finally, $(P_b X P_a)^* = P_a X^* P_b$. □

The multiplication law is ordinary matrix multiplication with operator-valued entries. It records the intermediate packet through which a composition passes. This matters even when two different paths reach the same destination. Their operator amplitudes add before any norm or probability is taken. Replacing each block by its norm would generally destroy interference and would no longer reconstruct the original operator.

Example 2.2 (A coarse block retains internal channels)**.** Use the two-packet split of Example 1.3 and the Hermitian matrix

$$H = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}.$$

Relative to $\mathbb{C}^4 = \mathbb{C}^2_A \oplus \mathbb{C}^2_B$, the four blocks are

$$H_{AA} = H_{BB} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad H_{BA} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}, \qquad H_{AB} = H_{BA}^*.$$

The block from A to B contains two distinct coordinate channels. Neither has to vanish for their net probability current to vanish at a particular state. The operator block describes available coupling. The current will also depend on the state occupying its source and destination.

Deleting off-diagonal blocks produces another operator, $\sum_a P_a X P_a$. It agrees with X exactly when all omitted blocks vanish. Thus an atlas resolution is lossless, while keeping only its diagonal is an additional operation. Similarly, a finite-dimensional coordinate choice inside the packet ranges turns the blocks into numerical submatrices, but that coordinate representation is not needed for reconstruction itself.

3 From skew-adjoint evolution to transfer current

Let $\psi(t)$ be a differentiable trajectory and suppose that

$$\dot\psi(t) = K(t)\psi(t), \qquad K(t)^* = -K(t). \tag{6}$$

Each $K(t)$ is bounded. The argument uses the stated derivative of the trajectory at the time under consideration. It does not manufacture a solution of an arbitrary differential equation. For a fixed bounded generator, the exponential series provides a separate construction of the evolution. A state-dependent vector field requires its own existence and uniqueness hypotheses.

Define the receiving-first current by

$$J_{b\leftarrow a}(t) = 2\,\mathrm{Re}\,\langle \psi_b(t), K(t)\psi_a(t)\rangle = 2\,\mathrm{Re}\,\langle \psi_b(t), K_{ba}(t)\psi_a(t)\rangle\,. \tag{7}$$

The second equality follows because $P_b\psi_b = \psi_b$. A positive value means that the contribution assigned to the pair increases the weight in b. The current is a real number, and its sign convention is fixed by the derivative calculation below.

Theorem 3.1 (Finite packet transfer)**.** *For the trajectory in* (6)*,*

$$\dot{p}_b = \sum_a J_{b\leftarrow a}, \tag{8}$$

$$J_{b\leftarrow a} = -J_{a\leftarrow b}, \qquad J_{a\leftarrow a} = 0, \tag{9}$$

$$\frac{d}{dt}\sum_a p_a = 0. \tag{10}$$

For every subset $C \subseteq I$, its total weight obeys

$$\frac{d}{dt}\sum_{b\in C} p_b = \sum_{b\in C}\sum_{a\notin C} J_{b\leftarrow a}. \tag{11}$$

Proof. The projectors are fixed in time, so the product rule for the inner product gives

$$\begin{aligned}\dot{p}_b &= 2\,\mathrm{Re}\left\langle P_b\psi, P_b\dot{\psi}\right\rangle = 2\,\mathrm{Re}\,\langle\psi_b, K\psi\rangle \\ &= \sum_a 2\,\mathrm{Re}\,\langle\psi_b, K\psi_a\rangle\,.\end{aligned}$$

The reconstruction of ψ justifies the last finite sum. Skew-adjointness gives

$$\langle\psi_a, K\psi_b\rangle = -\langle K\psi_a, \psi_b\rangle = -\overline{\langle\psi_b, K\psi_a\rangle}.$$

Taking real parts proves antisymmetry. Putting $a = b$ gives zero diagonal current. Every internal ordered pair in $C\times C$ cancels with its reverse. Splitting the sum over all sources into C and its complement proves (11). Taking $C = I$ proves total conservation. □

The first line of this proof does not require skew-adjointness. It expresses the derivative of the actual packet weight for any bounded generator. Skew-adjointness is the additional input that turns the resulting pairing into an antisymmetric transfer law. Without it, one generally has sources or sinks as well as exchange.

In the Hamiltonian convention $K = -iH$ with $H = H^*$ and $\hbar = 1$, linearity in the second slot yields

$$J_{b\leftarrow a} = 2\,\mathrm{Im}\,\langle\psi_b, H\psi_a\rangle\,. \tag{12}$$

Indeed, $\mathrm{Re}(-iz) = \mathrm{Im}\,z$. The plus sign belongs to this inner-product convention and to the choice that the first index receives the current. It should not be changed independently of either convention.

Example 3.2 (An exactly solved exchange)**.** On $\mathbb{C}^2$, take the coordinate atlas,

$$H = g\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad g > 0, \qquad \psi(0) = \begin{pmatrix}1\\ 0\end{pmatrix}.$$

Direct differentiation verifies the solution

$$\psi(t) = \begin{pmatrix} \cos(gt) \\ -i\sin(gt) \end{pmatrix}, \quad p_1 = \cos^2(gt), \quad p_2 = \sin^2(gt).$$

The current is

$$J_{2\leftarrow 1} = 2\,\mathrm{Im}(ig\sin(gt)\cos(gt)) = g\sin(2gt) = \dot{p}_2.$$

It reverses sign during the cycle. At $t = 0$ the coupling is nonzero but the instantaneous current is zero, since the destination amplitude vanishes. The population begins to change at second order in time. This is why coupling strength and instantaneous flow must be kept separate.

The same proof applies to $\dot{\psi} = K(\psi, t)\psi$ when the operator evaluated along the supplied trajectory is skew-adjoint. No derivative of the function $K(\psi, t)$ appears in the first derivative of $\|P_b\psi\|^2$. Such derivatives would enter a second derivative or a linearized perturbation equation. Pointwise conservation therefore survives many nonlinear choices of generator, while well-posedness remains a separate question.

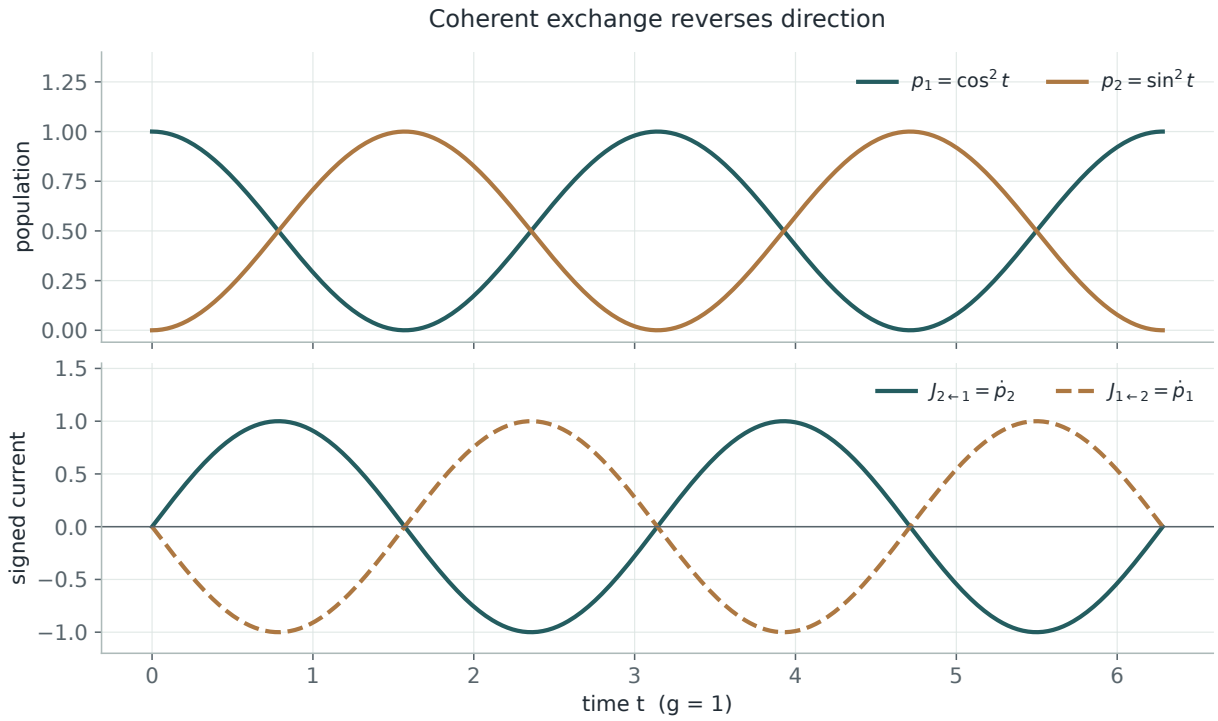


Figure 2. For the two-level Hamiltonian with coupling one, coherent evolution exchanges population periodically. The destination-first current changes sign when the direction of transfer reverses.

4 Coarsening preserves the balance exactly

Let $\mathcal{A}$ be a finite set of coarse labels and let $f : I \to \mathcal{A}$ assign every fine packet to exactly one coarse label. Write $C_A = f^{-1}(A)$ and define

$$Q_A = \sum_{a \in C_A} P_a. \tag{13}$$

The fibers are disjoint and cover I, so (Q_A) is another orthogonal atlas. Empty fibers give zero projectors. This is a grouping of an existing resolution. It need not be an approximation to a different physical system.

Theorem 4.1 (Exact coarse transfer)**.** *The coarse weight and current are*

$$p^{\mathrm{c}}_A = \|Q_A\psi\|^2 = \sum_{a\in C_A} p_a, \tag{14}$$

$$J^{\mathrm{c}}_{B\leftarrow A} = 2\,\mathrm{Re}\,\langle Q_B\psi, KQ_A\psi\rangle = \sum_{b\in C_B}\sum_{a\in C_A} J_{b\leftarrow a}. \tag{15}$$

They satisfy $\dot p^{\mathrm{c}}_B = \sum_A J^{\mathrm{c}}_{B\leftarrow A}$ and $J^{\mathrm{c}}_{B\leftarrow A} = -J^{\mathrm{c}}_{A\leftarrow B}$. In particular, the sum of fine currents internal to a coarse cell is zero.

Proof. Orthogonality proves the weight identity. Finite additivity of the real part of the inner product proves the current identity. The coarse projectors satisfy (1), so Theorem 3.1 applies to them. Equivalently, sum the fine balances inside each fiber and cancel all internal pairs. □

For an incidence matrix $R_{Aa} = 1$ when $f(a) = A$ and zero otherwise, these formulas read

$$p^{\mathrm{c}} = Rp, \qquad J^{\mathrm{c}} = RJR^{\mathsf{T}}, \qquad R^{\mathsf{T}}\mathbf{1}_{\mathrm{c}} = \mathbf{1}_{\mathrm{f}}. \tag{16}$$

The last identity expresses exactly one coarse assignment per fine index. An arbitrary binary matrix does not suffice. If S performs a second grouping, then

$$S(Rp) = (SR)p, \qquad S(RJR^{\mathsf{T}})S^{\mathsf{T}} = (SR)J(SR)^{\mathsf{T}}.$$

Thus a tower of exact groupings is associative. There is no accumulation of approximation error in these identities.

Example 4.2 (Internal exchange disappears from the coarse boundary)**.** Consider four labels with nonzero currents

$$J_{2\leftarrow 1} = 3, \quad J_{3\leftarrow 2} = 2, \quad J_{4\leftarrow 3} = 1, \quad J_{1\leftarrow 4} = \tfrac{1}{2},$$

their negatives in the reverse directions, and all other currents zero. This is an example of the balance algebra, without an asserted underlying quantum trajectory. The weight derivatives are $(-5/2, 1, 1, 1/2)$. For $A = \{1, 2\}$ and $B = \{3, 4\}$, the internal currents 3 and 1 cancel, leaving $J^{\mathrm{c}}_{B\leftarrow A} = 2 - 1/2 = 3/2$. Hence $\dot p^{\mathrm{c}}_A = -3/2$ and $\dot p^{\mathrm{c}}_B = 3/2$.

Exact balance should not be confused with closure of a reduced equation. The scalar vector p^{c} generally does not determine the state-dependent current J^{c}. Internal phase or correlation data may still be needed. Even at the fine level, a continuity equation alone does not uniquely determine its current. On three labels, the antisymmetric matrix

$$\begin{pmatrix} 0 & c & -c \\ -c & 0 & c \\ c & -c & 0 \end{pmatrix}$$

has zero row sums for any real c. Adding it changes circulation while preserving all weight derivatives. The operator and state specify a particular current through (7). Observing only the derivatives cannot recover that additional information.

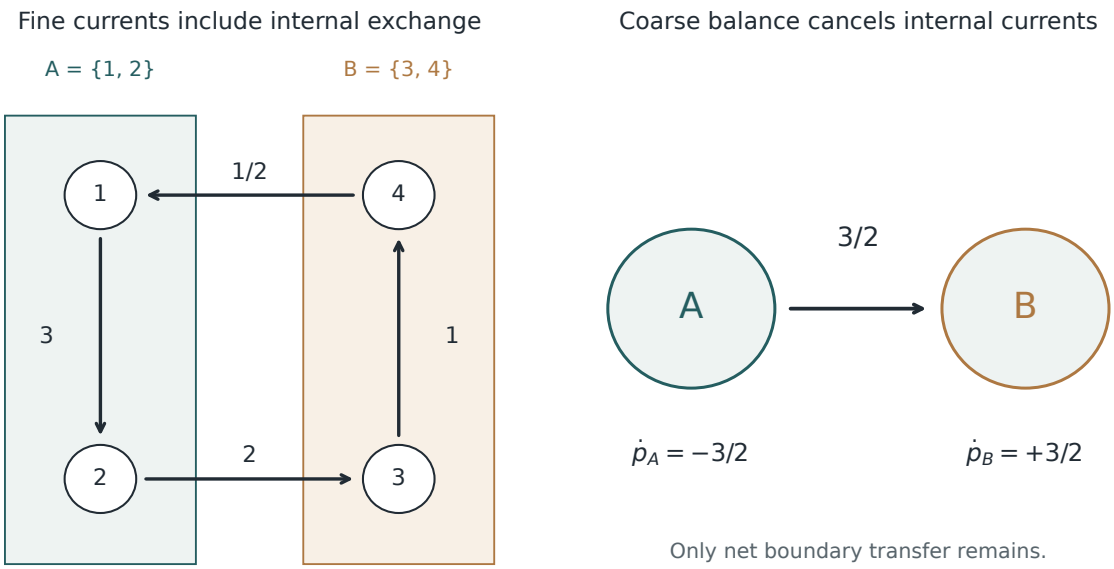


Figure 3. An instantaneous antisymmetric current array and its two-cell aggregation. The internal arrows cancel, leaving net transfer $3/2$ across the coarse boundary. This balance example does not assume a constructed Hamiltonian trajectory.

5 Moving atlases and their connection

A resolution can change with time even when the state is stationary. To retain orthogonality, transport the entire atlas by a unitary family. Let $U(t)$ be differentiable in operator norm and satisfy both $U^*U = \mathrm{id}$ and $UU^* = \mathrm{id}$. Set

$$P_a(t) = U(t)P_a^{(0)}U(t)^*, \qquad \phi(t) = U(t)^*\psi(t), \qquad \Gamma(t) = U(t)^*\dot{U}(t). \tag{17}$$

The coordinates ϕ see fixed projectors $P_a^{(0)}$. The physical weight in the moving packet is the same scalar:

$$\|P_a(t)\psi(t)\|^2 = \left\|P_a^{(0)}\phi(t)\right\|^2.$$

Proposition 5.1 (The transformed generator)**.** *If $\dot{\psi} = K\psi$ and $K^* = -K$, then*

$$\dot{\phi} = K_{\mathrm{eff}}\phi, \qquad K_{\mathrm{eff}} = U^*KU - \Gamma, \qquad \Gamma^* = -\Gamma, \quad K_{\mathrm{eff}}^* = -K_{\mathrm{eff}}. \tag{18}$$

Consequently the fixed-atlas current formula, applied to ϕ and K_{eff}, gives the derivative of the moving physical packet weights.

Proof. Differentiate $U^*U = \mathrm{id}$ to obtain $\dot{U}^*U + U^*\dot{U} = 0$. This proves $\Gamma^* = -\Gamma$. Multiplication on the right by U^* gives $\dot{U}^* = -\Gamma U^*$. Therefore

$$\dot{\phi} = \dot{U}^*\psi + U^*\dot{\psi} = -\Gamma\phi + U^*KU\phi.$$

Unitary conjugation preserves skew-adjointness. The difference of the two skew-adjoint operators in (18) is skew-adjoint, so the transfer theorem applies. □

The connection Γ records the velocity of the frame. Omitting it would interpret a change of packet coordinates as if it were entirely a change of the physical state. In Hamiltonian notation the corresponding effective Hamiltonian is $H_{\text{eff}} = U^* H U - i\Gamma$, which is self-adjoint because Γ is skew-adjoint.

Example 5.2 (A stationary vector viewed through a rotating split)**.** Let $\psi(t) = e_1$, $K = 0$, and

$$U(t) = \begin{pmatrix} \cos\theta(t) & -\sin\theta(t) \\ \sin\theta(t) & \cos\theta(t) \end{pmatrix}.$$

With the coordinate projectors as $P_a^{(0)}$,

$$\phi = \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix}, \quad \Gamma = \dot\theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad K_{\text{eff}} = -\Gamma.$$

The moving weights are $p_1 = \cos^2\theta$ and $p_2 = \sin^2\theta$, with $J_{2\leftarrow 1} = \dot\theta \sin(2\theta) = \dot p_2$. The state has not moved in the original Hilbert space. The measured sectors have moved around it. The connection term accounts for the entire change.

The same principle governs a time-dependent operator block. Write $V = \dot U U^*$ for the frame velocity in laboratory coordinates. Then

$$\dot P_a = [V, P_a], \qquad \frac{d}{dt}(P_b X P_a) = [V, P_b] X P_a + P_b \dot X P_a + P_b X [V, P_a].$$

Both packet boundaries contribute, as does the operator itself. Conjugation by a general invertible family also preserves idempotence and completeness, but it need not preserve self-adjointness or orthogonality. The unitary hypothesis is what keeps these transported objects within the present orthogonal atlas setting.

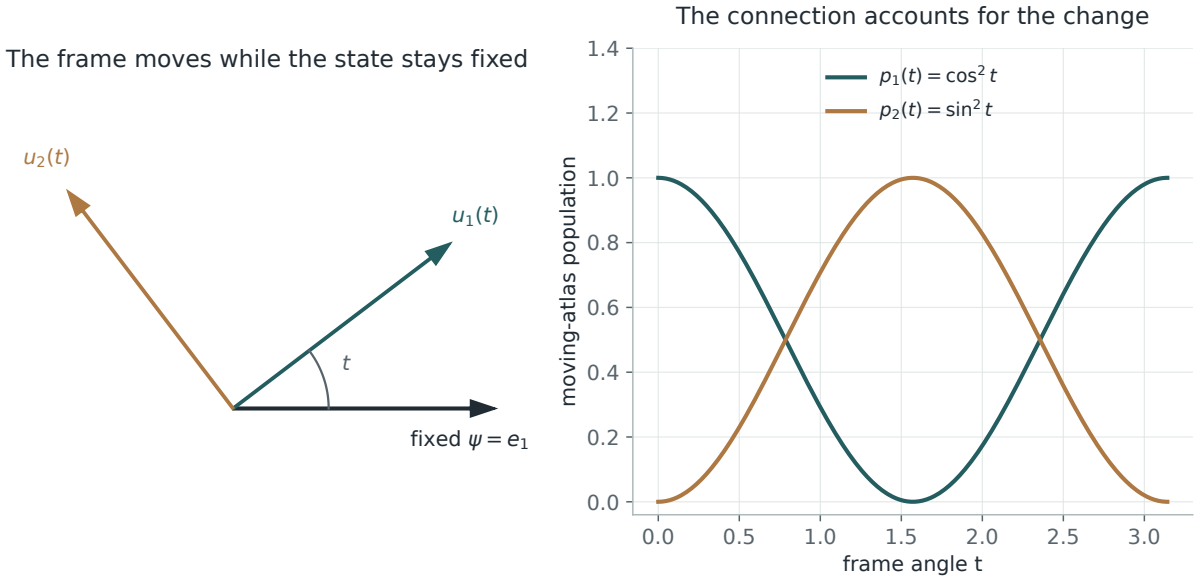


Figure 4. A stationary laboratory vector observed in a rotating atlas. The varying packet weights are generated by the connection term $-U^*\dot U$ in moving coordinates.

6 Mixed states and coherent transfer

For this section take $\mathcal{H} = \mathbb{C}^n$. A density matrix is a positive semidefinite Hermitian matrix ρ with $\operatorname{Tr}\rho = 1$. Its probability in a sharp packet is

$$p_a = \operatorname{Tr}(P_a\rho). \tag{19}$$

Hermiticity makes this trace real. Positivity makes it nonnegative, since cyclicity and $P_a^2 = P_a$ give $\operatorname{Tr}(P_a\rho) = \operatorname{Tr}(P_a\rho P_a) \geq 0$. Completeness gives $\sum_a p_a = \operatorname{Tr}\rho = 1$. These are separate uses of the density-matrix assumptions.

Suppose a differentiable matrix trajectory obeys the von Neumann equation

$$\dot\rho = -i[H,\rho], \qquad H = H^*. \tag{20}$$

For its coherent transfer define

$$J^{(\rho)}_{b\leftarrow a} = 2\operatorname{Im}\operatorname{Tr}(P_bHP_a\rho). \tag{21}$$

Proposition 6.1 (Trace-current balance)**.** *For a Hermitian ρ satisfying* (20)*,*

$$\dot p_b = \sum_a J^{(\rho)}_{b\leftarrow a}, \qquad J^{(\rho)}_{b\leftarrow a} = -J^{(\rho)}_{a\leftarrow b}.$$

The subset boundary formula and exact coarse-current summation remain valid.

Proof. Let $z_{ba} = \operatorname{Tr}(P_bHP_a\rho)$. Taking the conjugate trace and cyclically moving the first factor to the end yields

$$\overline{z_{ba}} = \operatorname{Tr}(\rho P_aHP_b) = \operatorname{Tr}(P_aHP_b\rho) = z_{ab}.$$

Completeness and cyclicity resolve the derivative as

$$\dot p_b = -i\sum_a (z_{ba} - z_{ab}) = \sum_a 2\operatorname{Im} z_{ba}.$$

The conjugate relation gives antisymmetry. Finite sums then prove cancellation and coarse aggregation exactly as before. □

For a rank-one state $\rho = |\psi\rangle\langle\psi|$, where the associated operator sends x to $\psi\,\langle\psi, x\rangle$, one has $\operatorname{Tr}(M\rho) = \langle\psi, M\psi\rangle$. Substituting $M = P_bHP_a$ proves that (21) is precisely the pure-state formula. No normalization is needed for that identity itself.

In the coordinate atlas, the current is $2\operatorname{Im}(H_{ba}\rho_{ab})$. It depends on off-diagonal coherence. A useful test is to compare

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}, \qquad \rho_- = \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}, \qquad \rho_0 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

All three are density matrices with packet probabilities $(1/2, 1/2)$. For $H = g\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, however, their currents $J_{2\leftarrow 1}$ are respectively $g, -g, 0$. The same instantaneous populations can therefore have different derivatives under the same Hamiltonian. A closed equation involving only those two populations cannot represent all three coherent states.

The trace balance requires Hermiticity, but its algebra does not require positivity or trace one. Those additional hypotheses justify the probabilistic interpretation. This distinction is useful in formalization because it reveals which assumptions belong to a calculation and which belong to interpreting its output. Infinite-dimensional mixed states would require a trace-class setting and further control of products and derivatives. They are not silently included in this finite matrix argument.

7 Tensor packets, marginals, and retained correlations

Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be finite-dimensional Hilbert spaces with finite orthogonal atlases (P_a) and (Q_μ). Their tensor atlas is

$$R_{a\mu} = P_a \otimes Q_\mu.$$

It is orthogonal because tensor multiplication gives $R_{a\mu}R_{b\nu} = \delta_{ab}\delta_{\mu\nu}R_{a\mu}$. Its sum is $\mathrm{id}_A \otimes \mathrm{id}_B$. For a density matrix ρ_{AB}, define

$$p_{a\mu} = \mathrm{Tr}(R_{a\mu}\rho_{AB}), \qquad p^A_a = \sum_\mu p_{a\mu}, \quad p^B_\mu = \sum_a p_{a\mu}. \tag{22}$$

These marginalizations are exact atlas coarsenings. For example,

$$p^A_a = \mathrm{Tr}((P_a \otimes \mathrm{id}_B)\rho_{AB}) = \mathrm{Tr}(P_a\rho_A), \qquad \rho_A = \mathrm{Tr}_B\,\rho_{AB}.$$

Choose full orthonormal bases of the two Hilbert spaces, with $(e_k)_k$ indexing the basis of $\mathcal{H}_B$. In the resulting product basis the partial trace has entries $(\rho_A)_{ij} = \sum_k (\rho_{AB})_{(i,k),(j,k)}$. This explicit formula shows that it preserves Hermiticity and trace. Positivity follows by evaluating the quadratic form on x and summing the nonnegative forms of $x \otimes e_k$. Thus the reduced matrix is itself a density matrix.

For a product state $\rho_{AB} = \sigma_A \otimes \tau_B$, the trace factorizes and $p_{a\mu} = p^A_a p^B_\mu$. General states need not have that property. The normalized vector $\Psi = (e_0 \otimes e_0 + e_1 \otimes e_1)/\sqrt{2}$ has joint probabilities

$$(p_{a\mu}) = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad p^A = p^B = (1/2, 1/2).$$

The product density matrix $(\mathrm{id}_A/2) \otimes (\mathrm{id}_B/2)$ has the same marginals but every joint entry is $1/4$. Marginals therefore fail to reconstruct even the joint measurement distribution. Moreover, the classical mixture of $e_0 \otimes e_0$ and $e_1 \otimes e_1$ has the same joint table as Ψ but lacks its off-diagonal coherence. A single atlas table does not reconstruct the full density matrix or certify entanglement on its own.

When a joint coherent current is defined on the product labels, the current into a marginal packet is obtained by summing over the hidden labels on both its source and destination. Internal exchange within a marginal cell cancels. Correlations can still enter the surviving current, so exact marginal balance does not imply autonomous marginal dynamics. The distinction repeats the one already encountered in coarse pure-state transfer.

8 Open dynamics and rank-one jump flow

Coherent exchange is one mechanism for changing packet weights. A second comes from an explicitly chosen family of dissipative transitions. Fix an orthonormal basis $(e_a)_{a\in I}$ of $\mathbb{C}^n$, use its rank-one projectors, and choose rates $\gamma_{ba} \geq 0$ for $b \neq a$. Set diagonal rates to zero for convenience and define

$$L_{ba} = \sqrt{\gamma_{ba}}\,|e_b\rangle\langle e_a|, \qquad \mathcal{D}_L(\rho) = L\rho L^* - frac12(L^*L\rho + \rho L^*L). \tag{23}$$

Consider a supplied differentiable matrix trajectory satisfying

$$\dot\rho = \sum_{b\neq a} \mathcal{D}_{L_{ba}}(\rho). \tag{24}$$

This choice of jumps is tied to the selected basis. A general dissipative operator or a higher-rank atlas does not automatically produce the same closed rate equation.

Proposition 8.1 (Population gain and loss)**.** *The diagonal populations $p_b = \rho_{bb}$ of a Hermitian solution satisfy*

$$\dot p_b = \sum_a F_{b\leftarrow a}, \qquad F_{b\leftarrow a} = \gamma_{ba}p_a - \gamma_{ab}p_b, \qquad F_{b\leftarrow a} = -F_{a\leftarrow b}. \tag{25}$$

The total population is constant, and every coarse population has the sum of the boundary jump currents as its derivative.

Proof. For the one transition $a \to b$,

$$L_{ba}\rho L_{ba}^* = \gamma_{ba}p_aP_b, \qquad L_{ba}^*L_{ba} = \gamma_{ba}P_a.$$

Its diagonal gain is $\gamma_{ba}p_a$ at b, and the two halves of the anticommutator remove that same amount at a. No other diagonal entry changes from this one jump. Sum over ordered transitions to obtain (25). Antisymmetry is immediate, and the finite boundary cancellation applies. □

In this model the population equation closes without knowing the coherences. The coherences nevertheless have their own dynamics. For $i \neq j$, let $r_i = \sum_{b\neq i}\gamma_{bi}$ be the total rate leaving i. The gain matrices are diagonal, while each anticommutator selects its source row and column. Consequently

$$\dot\rho_{ij} = -\tfrac{1}{2}(r_i + r_j)\rho_{ij}. \tag{26}$$

For fixed rates, $\rho_{ij}(t) = e^{-(r_i+r_j)t/2}\rho_{ij}(0)$. This decay law belongs to the stated rank-one jump family. Additional Hamiltonian or dephasing terms alter it in identifiable ways.

Example 8.2 (Two-state relaxation)**.** Take $\gamma_{21} = \gamma_{12} = k > 0$ and $p_1 + p_2 = 1$. Then

$$\dot p_1 = k(1-2p_1), \qquad p_1(t) = \tfrac{1}{2} + (p_1(0) - \tfrac{1}{2})e^{-2kt}, \qquad \rho_{12}(t) = e^{-kt}\rho_{12}(0).$$

The populations relax toward equal weights. In the coherent exchange of Example 3.2, the populations oscillate and depend on phase. Both systems admit antisymmetric probability currents, but that common bookkeeping does not make their mechanisms identical. Adding $-i[H,\rho]$ to (24) adds the coherent current to the jump current. The resulting population equation then generally depends on coherence again.

The coarse jump balance is always exact. For it to close as a time-homogeneous Markov equation for coarse weights alone, a further compatibility condition suffices. For distinct coarse cells A, B, suppose

$$\sum_{b \in C_B} \gamma_{ba} = q_{BA} \quad \text{for every } a \in C_A.$$

Then summing the fine gains gives $q_{BA} p^{\mathrm{c}}_A$, so $\dot{p}^{\mathrm{c}}_B = \sum_{A \neq B} (q_{BA} p^{\mathrm{c}}_A - q_{AB} p^{\mathrm{c}}_B)$. Without this equality of outgoing rates to each destination cell, the fine population distribution inside a source cell matters. This direct consequence of the gain sum shows precisely what an autonomous coarse model asks in addition to current conservation.

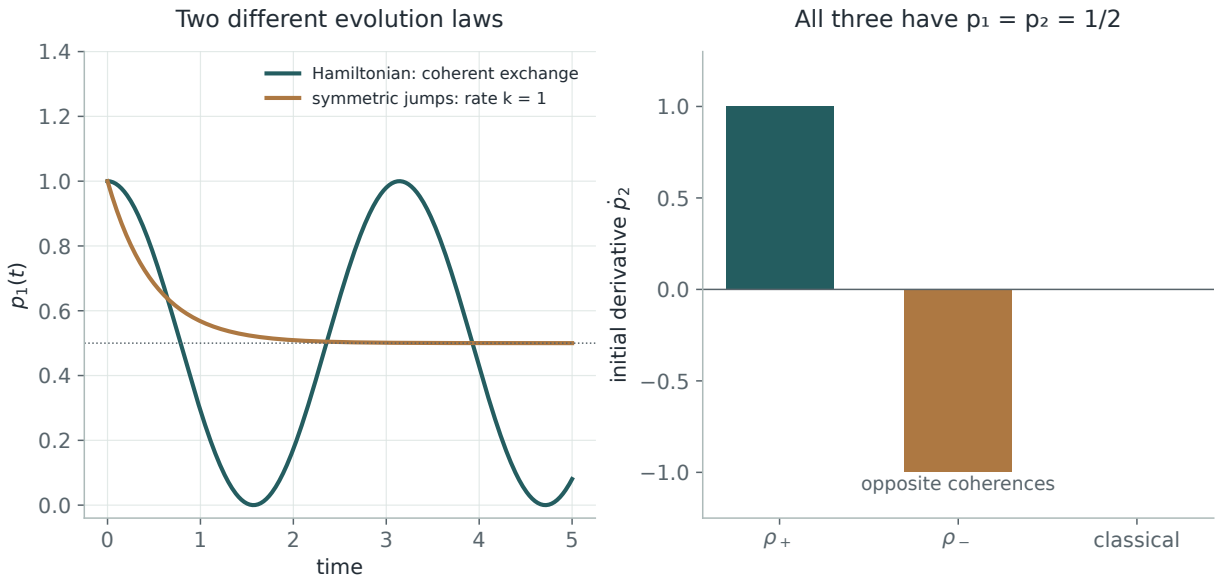


Figure 5. Coherent exchange and the specified symmetric rank-one jump dynamics have different population laws. The equal-population coherent states ρ_+ and ρ_- give opposite initial derivatives of p_2, while the classical jump equilibrium has zero derivative.

9 Information functionals and the role of detailed balance

An antisymmetric current also organizes derivatives of scalar functionals of the packet distribution. Let $p_a(t) > 0$ be differentiable, normalized, and obey $\dot{p}_b = \sum_a J_{b \leftarrow a}$ with J antisymmetric. For a differentiable real function f on the positive half-line,

$$\frac{d}{dt} \sum_b f(p_b) = \frac{1}{2} \sum_{b,a} J_{b \leftarrow a} (f'(p_b) - f'(p_a)). \tag{27}$$

To prove it, apply the ordinary chain rule and substitute the balance equation. Exchange the two indices in a second copy of the sum and use antisymmetry. Averaging the two expressions yields (27).

For Shannon entropy $S(p) = -\sum_a p_a \log p_a$, this gives

$$\dot{S} = -\frac{1}{2} \sum_{b,a} J_{b \leftarrow a} \log \frac{p_b}{p_a}. \tag{28}$$

The constant derivative term cancels because the total current is zero. For a real q, the power sum $M_q = \sum_a p_a^q$ has derivative

$$\dot{M}_q = \frac{q}{2} \sum_{b,a} J_{b\leftarrow a}(p_b^{q-1} - p_a^{q-1}).$$

For $q \neq 1$, the Rényi entropy $H_q = \log M_q/(1-q)$ therefore satisfies $\dot{H}_q = \dot{M}_q/((1-q)M_q)$. Strict positivity makes the powers and logarithmic derivatives valid, and $M_q > 0$ when the finite index set is nonempty. Boundary points with zero populations require their own limiting argument and are not covered by these derivative statements as written.

These formulas alone impose no sign on entropy change. The coherent two-state orbit can increase and then decrease its packet entropy as the state moves through a cycle. Packet entropy depends on the chosen atlas, and it is different from the von Neumann entropy of the density operator. In particular, a pure state can have positive packet entropy.

For the rank-one jump model, detailed balance supplies a definite monotonicity theorem. Let $\pi_a > 0$ be a fixed probability distribution, and assume

$$\gamma_{ba}\pi_a = \gamma_{ab}\pi_b. \tag{29}$$

Define relative densities $u_a = p_a/\pi_a$ and conductances $c_{ba} = \gamma_{ba}\pi_a$. Then $c_{ba} = c_{ab} \geq 0$ and $F_{b\leftarrow a} = c_{ba}(u_a - u_b)$.

Theorem 9.1 (Relative-entropy dissipation)**.** *For a strictly positive solution of the finite jump balance and a fixed positive reference satisfying* (29)*,*

$$D(p\|\pi) = \sum_a p_a \log \frac{p_a}{\pi_a}, \tag{30}$$
$$\frac{d}{dt} D(p\|\pi) = -\frac{1}{2} \sum_{b,a} c_{ba}(u_b - u_a)(\log u_b - \log u_a) \leq 0.$$

Proof. Differentiate with π fixed. The terms without logarithms sum to zero. Pairing opposite currents leaves $\frac{1}{2}\sum_{b,a} F_{b\leftarrow a}(\log u_b - \log u_a)$. Insert the conductance expression for F. The logarithm is increasing on $(0,\infty)$, so each product $(u_b - u_a)(\log u_b - \log u_a)$ is nonnegative. Every conductance is nonnegative, proving the inequality. □

The conductance form includes inactive edges without evaluating a logarithm of a zero flux. When $c_{ba} = 0$, its contribution is simply zero. When $c_{ba} > 0$, a summand vanishes exactly when $u_b = u_a$. Thus zero instantaneous dissipation means that the relative density is constant on each connected component of the positive-conductance graph. If that graph is connected, normalization forces the constant to be one and $p = \pi$. This identifies the equality case. The derivative calculation by itself does not provide a convergence rate to equilibrium.

The finite calculus now has a common structure across pure states, mixed states, moving coordinates, tensor marginals, and the specified jump model. Each setting begins with actual mathematical objects and a stated derivative. Reconstruction produces the balance, antisymmetry gives cancellation, and an exact grouping preserves the resulting current. Extending this chain to positive effects, countable families, or unbounded generators requires new hypotheses at the step where the original finite proof used orthogonality, finite summation, or bounded application.

10 Refinement cancellation and the cost of extrapolation

Coarsening a packet atlas combines exact components of one description. A different use of refinement compares approximations made at several scales. The distinction matters. An algebraic cancellation of leading error terms requires an expansion in a common vector space, and a useful numerical result also requires control of the remaining error and of the sample perturbations.

Let E be a real normed vector space and let $f(h) \in E$ be defined for $0 < |h| < \delta$. Fix a refinement ratio $b > 1$. If $f(h)$ represents operators on different Hilbert spaces, comparison maps must first place them in a common space where the following additions are meaningful. The formulas alone do not construct those maps.

The denominators below are positive. Each level is a finite linear combination of the original samples at successively refined scales.

Definition 10.1 (The actual refinement tower)**.** Set $T_0 f(h) = f(h)$ and, for $m \geq 0$, define

$$T_{m+1} f(h) = \frac{b^{m+1} T_m f(h/b) - T_m f(h)}{b^{m+1} - 1}.$$

Proposition 10.2 (Exact response of a monomial)**.** *For $r \in \mathbb{N}$ and $v \in E$,*

$$T_m(h^r v) = c_{m,r} h^r v, \qquad c_{m,r} = \prod_{k=1}^{m} \frac{b^{k-r} - 1}{b^k - 1}.$$

The empty product equals one. In particular, $c_{m,0} = 1$, and $c_{m,r} = 0$ for $1 \leq r \leq m$.

Proof. At level zero the assertion is the definition. If it holds at level m, the next recursion multiplies the coefficient by

$$\frac{b^{m+1} b^{-r} - 1}{b^{m+1} - 1}.$$

This appends the next factor to the product. For $r = 0$, each factor equals one. For $1 \leq r \leq m$, the factor at $k = r$ vanishes. □

Suppose that a finite expansion is available,

$$f(h) = F + \sum_{r=1}^{m} h^r v_r + R(h).$$

Linearity and the proposition give the exact identity

$$T_m f(h) - F = T_m R(h).$$

Only a finite sum has been used. No convergence of an infinite formal series is needed. The size of the right-hand side remains an analytic question.

Theorem 10.3 (A quantified remainder and perturbation bound)**.** *Let p be a nonnegative integer and suppose*

$$\|R(h)\| \le C|h|^p, \qquad \|g(h) - f(h)\| \le \varepsilon \quad (0 < |h| < \delta),$$

where $C, \varepsilon \ge 0$. Define

$$A_{m,p} = \prod_{k=1}^{m} \frac{b^{k-p} + 1}{b^k - 1}.$$

Then, throughout the same punctured interval,

$$\|T_m g(h) - F\| \le A_{m,p} C|h|^p + A_{m,0}\varepsilon.$$

Proof. The interval is stable under $h \mapsto h/b$. The triangle inequality applied to one tower step gives

$$\|T_{m+1}R(h)\| \le \frac{b^{m+1}\,\|T_m R(h/b)\| + \|T_m R(h)\|}{b^{m+1} - 1}.$$

Starting with the stated remainder bound and proceeding by induction multiplies its constant at step $m+1$ by $(b^{m+1-p}+1)/(b^{m+1}-1)$. Hence $\|T_m R(h)\| \le A_{m,p}C|h|^p$. Apply the same argument with $p = 0$ to $g - f$. Linearity and the triangle inequality separate the two contributions. □

When $p = m+1$, this proves the expected order improvement under an explicit remainder hypothesis. It also retains the effect of noisy samples. Each factor in $A_{m,0}$ exceeds one, so extrapolation can amplify a uniform sample error even while cancelling several deterministic powers of h.

10.1 A complete three-level example

For $b = 2$, direct substitution gives

$$\begin{aligned} T_1 f(h) &= 2f(h/2) - f(h), \\ T_2 f(h) &= \frac{8f(h/4) - 6f(h/2) + f(h)}{3}, \\ T_3 f(h) &= \frac{64f(h/8) - 56f(h/4) + 14f(h/2) - f(h)}{21}. \end{aligned}$$

Take the exact scalar polynomial $f(h) = 1 + 2h - 3h^2 + 4h^3$. The responses are

$$T_1 f(h) = 1 + \frac{3}{2}h^2 - 3h^3, \qquad T_2 f(h) = 1 + \frac{1}{2}h^3, \qquad T_3 f(h) = 1.$$

These are polynomial identities for every real h. If each of the three samples used by T_2 has absolute error at most ε, its output error is at most

$$\frac{8 + 6 + 1}{3}\varepsilon = 5\varepsilon.$$

This bound is attained by errors with signs matching the three weights. Thus exact removal of the first two powers does not imply a contraction on arbitrary perturbations. For the displayed polynomial, the perturbed second-level approximation has the explicit total bound $|h|^3/2 + 5\varepsilon$.

The calculation supplies a practical reading rule for later refinement claims. An exact coefficient identity, an asymptotic remainder estimate, and a stability bound are separate pieces of evidence. A

physical approximation uses all three, together with the comparison maps that make its values commensurable.

Limits, domains, and differential structure

11 Positive effects and sharp dilation

A sharp packet atlas assigns mutually orthogonal subspaces to its labels. Many useful descriptions overlap. A detector may respond partly to several channels, and a frame may analyze the same component more than once. The correct replacement is a positive partition of the identity. Its weights are expectation values, and its channel vectors live in a larger space where orthogonality can be restored.

Throughout this chapter, inner products are conjugate linear in the first variable. A bounded state generator is denoted by K, with $K^* = -K$. When a Hamiltonian is present, $K = -iH$. The receiving label is written first in J_{ba}.

Definition 11.1 (Finite effect atlas)**.** Let $\mathcal{H}$ be a complex Hilbert space and let A be a finite label set. An effect atlas is a family of bounded operators satisfying

$$F_a = F_a^* \geq 0, \qquad \sum_{a \in A} F_a = I.$$

Its weight in a state ψ is $p_a = \langle \psi, F_a \psi \rangle$. The partition identity implies $0 \leq F_a \leq I$, so $p_a \geq 0$ and $\sum_a p_a = \|\psi\|^2$.

For a sharp atlas, $F_a = P_a$ and $p_a = \|P_a \psi\|^2$. In general, p_a is not $\|F_a \psi\|^2$. The latter quantity is $\langle \psi, F_a^2 \psi \rangle$ and therefore loses precisely the positive defect $F_a - F_a^2$.

Example 11.2 (Overlapping two-channel weights)**.** On $\mathbb{C}^2$, take

$$F_+ = \begin{pmatrix} 4/5 & 0 \\ 0 & 1/5 \end{pmatrix}, \qquad F_- = \begin{pmatrix} 1/5 & 0 \\ 0 & 4/5 \end{pmatrix}, \qquad \psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The true weights are $4/5$ and $1/5$. Squared effect-vector norms would instead give $16/25$ and $1/25$, whose sum is $17/25$. The state still satisfies $F_+ \psi + F_- \psi = \psi$, but its two summands are not orthogonal. Reconstruction and an orthogonal decomposition are different assertions.

11.1 Continuity without orthogonality

Suppose that a differentiable state curve satisfies $\dot{\psi} = K\psi$. Define

$$J_{ba}^F = \operatorname{Re} \langle F_b \psi, K F_a \psi \rangle - \operatorname{Re} \langle F_a \psi, K F_b \psi \rangle . \tag{31}$$

This current is antisymmetric directly from its definition. Skew-adjointness also gives

$$J_{ba}^F = 2 \operatorname{Re} \langle F_b \psi, K F_a \psi \rangle .$$

The same bilinear expression occurs for sharp packets, but its interpretation now concerns expectation weights.

Proposition 11.3 (Effect continuity)**.** *Under these hypotheses,*

$$\dot{p}_b = \sum_a J^F_{ba}, \qquad \sum_b \dot{p}_b = 0.$$

If $F_b = F_b(t)$ is differentiable in operator norm, the first identity becomes

$$\dot{p}_b = s_b + \sum_a J^F_{ba}, \qquad s_b = \langle \psi, \dot{F}_b \psi \rangle , \qquad \sum_b s_b = 0.$$

Proof. Self-adjointness of F_b and differentiation give

$$\dot{p}_b = 2 \operatorname{Re} \langle F_b \psi, K\psi \rangle$$

for a fixed effect. Summing the two terms of (31) over a and using $\sum_a F_a = I$ gives

$$\sum_a J^F_{ba} = \operatorname{Re} \langle F_b \psi, K\psi \rangle - \operatorname{Re} \langle \psi, K F_b \psi \rangle .$$

The second real part is the negative of the first. The moving case adds the product-rule term s_b. Its total is zero because $\sum_b \dot{F}_b = 0$. Antisymmetry cancels the finite double sum of currents. □

If a map $r : A \to B$ groups the labels, the coarse effect is $G_\beta = \sum_{r(a)=\beta} F_a$. Bilinearity proves

$$p^{\mathrm{c}}_\beta = \sum_{r(a)=\beta} p_a, \qquad J^{\mathrm{c}}_{\beta\alpha} = \sum_{\substack{r(b)=\beta \\ r(a)=\alpha}} J^F_{ba}.$$

No orthogonality is needed for this finite aggregation law. A divergence-free antisymmetric array may also be added to J^F without changing its row sums. Thus the balance law determines the change in weight, while the generator-induced formula selects a particular representative of the current.

11.2 Analysis, synthesis, and the range projection

Choose Hilbert spaces $\mathcal{K}_a$ and bounded maps $V_a : \mathcal{H} \to \mathcal{K}_a$ such that

$$\sum_a V_a^* V_a = I.$$

The combined analysis map $V : \mathcal{H} \to \mathcal{K} = \bigoplus_a \mathcal{K}_a$ is $V\psi = (V_a\psi)_a$. Its adjoint is synthesis, $V^*(y_a)_a = \sum_a V_a^* y_a$. Consequently

$$V^*V = I, \qquad R = VV^*, \qquad R^* = R^2 = R.$$

Here R projects onto the closed range of the isometry V. The range need not fill $\mathcal{K}$. If Π_a is the coordinate projector, the compressed effect is

$$F_a = V^* \Pi_a V = V_a^* V_a, \qquad p_a = \|V_a\psi\|^2 = \|\Pi_a V\psi\|^2 .$$

Every finite effect atlas admits this construction by taking $\mathcal{K}_a = \mathcal{H}$ and $V_a = F_a^{1/2}$. Positivity supplies the bounded positive square root. No choice of a measurement outcome or post-measurement state is being made by this algebraic dilation.

For $X \in \mathcal{B}(\mathcal{H})$, its channel block is $X^V_{ba} = V_b X V_a^*$. Inserting the partition on both sides proves

$$X = \sum_{a,b} V_b^* X^V_{ba} V_a. \tag{32}$$

The lifted operator $\widetilde{X} = VXV^*$ is multiplicative on lifted operators, since $V^*V = I$. Its identity is R, not $I_{\mathcal{K}}$. Conversely, compressing arbitrary ambient operators need not preserve multiplication:

$$V^*ABV - (V^*AV)(V^*BV) = V^*A(I-R)BV. \tag{33}$$

This is the exact missing-range term. Setting $A = B = \Pi_a$ gives

$$\langle \psi, (F_a - F_a^2)\psi \rangle = \left\| (I-R)\Pi_a V \psi \right\|^2. \tag{34}$$

Thus the failure of an effect to be sharp measures how much its coordinate projection leaves the embedded physical range. This identity also proves $F_a - F_a^2 \geq 0$ directly.

The dynamics respects the same construction. Put $\widetilde{K} = VKV^*$. It is skew-adjoint, and $\frac{d}{dt} V\psi = \widetilde{K} V\psi$. Moving adjoints across the inner product shows

$$2\,\mathrm{Re}\left\langle \Pi_b V\psi, \widetilde{K}\Pi_a V\psi \right\rangle = 2\,\mathrm{Re}\,\langle F_b\psi, KF_a\psi\rangle = J^F_{ba}.$$

The positive-effect current is therefore exactly the sharp current of this lifted flow evaluated on the embedded state. It is not obtained by treating arbitrary compression as an algebra homomorphism.

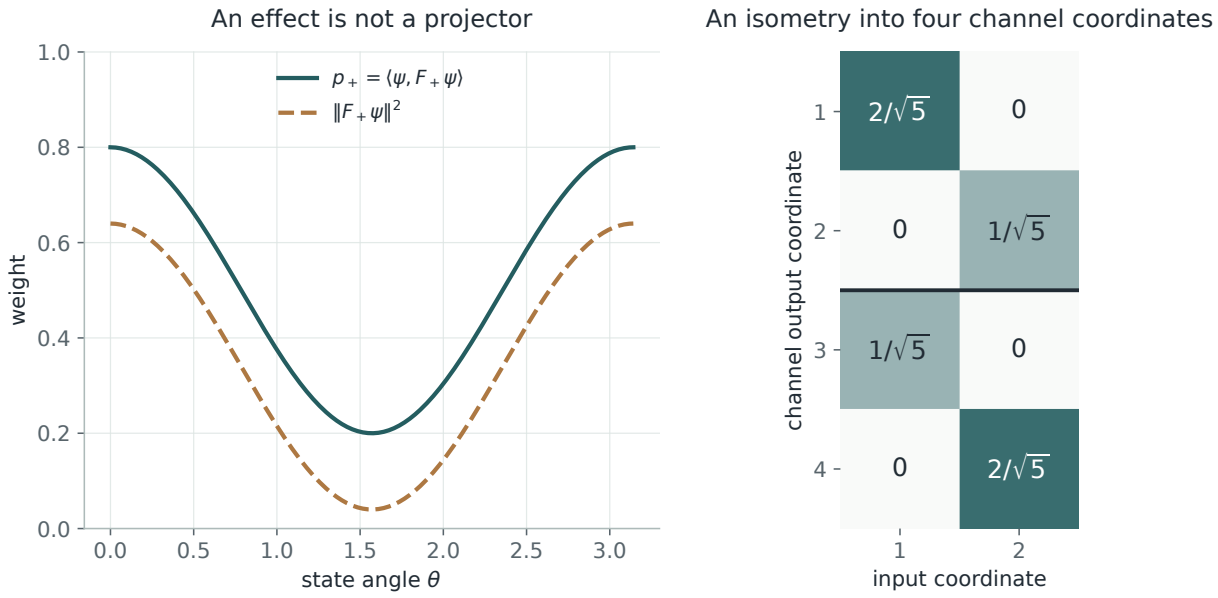


Figure 6. An explicit two-effect example with $F_+ = \mathrm{diag}(4/5, 1/5)$ and $F_- = I - F_+$. The square-root channel isometry preserves the effect probabilities in an enlarged sharp resolution.

12 Countable packets and the domain of a generator

An infinite resolution introduces two separate limits. Packet vectors must reconstruct the state, and applying an unbounded operator to the partial reconstruction requires additional control. Writing an infinite sum of matrix blocks conceals this distinction unless the domain is stated first.

Let $(P_a)_{a\in A}$ be a countable orthogonal family of bounded projections with strong sum I. For finite $F \subset A$, write $S_F = \sum_{a\in F} P_a$. Strong completeness means $S_F x \to x$ for each fixed $x \in \mathcal{H}$, over the directed set of finite subsets. Orthogonality gives

$$\|S_F x\|^2 = \sum_{a\in F} \|P_a x\|^2, \qquad \sum_a \|P_a x\|^2 = \|x\|^2.$$

These are Hilbert-space statements. They contain no unbounded operator.

Now let $H : D(H) \subset \mathcal{H} \to \mathcal{H}$ be a densely defined symmetric operator. To write $HP_a\psi$ for every domain state, require

$$P_a D(H) \subset D(H). \tag{35}$$

Define $\|x\|_H^2 = \|x\|^2 + \|Hx\|^2$. A domain-compatible atlas is graph compatible when

$$S_F x \longrightarrow x \quad \text{in the graph norm for every } x \in D(H). \tag{36}$$

Equivalently, both $S_F x \to x$ and $HS_F x \to Hx$ hold in $\mathcal{H}$. The second convergence does not follow from the first for an unbounded H.

Example 12.1 (A small state can have a large generator image)**.** On $\ell^2(\mathbb{N}_{\geq 1})$, let $He_n = ne_n$ on its maximal domain. The finitely supported vectors $x_N = e_N/N$ satisfy $\|x_N\| = 1/N \to 0$, while $\|Hx_N\| = 1$. This gives a direct reason to track the generator tail in addition to the state tail. It does not contradict closedness, because $Hx_N = e_N$ has no norm limit.

12.1 A maximal diagonal domain

Let (λ_n) be any real sequence and define

$$D(H_\lambda) = \left\{ c \in \ell^2 : \sum_n \lambda_n^2 |c_n|^2 < \infty \right\}, \qquad (H_\lambda c)_n = \lambda_n c_n.$$

The operator is dense because it contains every finitely supported sequence. It is symmetric by the reality of the multipliers. Its domain is maximal for this coordinate action.

Proposition 12.2 (Self-adjointness of the diagonal realization)**.** *The operator H_λ is self-adjoint. Its coordinate cutoffs converge in graph norm on $D(H_\lambda)$.*

Proof. If $y \in D(H_\lambda^*)$ and $H_\lambda^* y = z$, testing the adjoint identity against e_n gives $z_n = \lambda_n y_n$. Since $z \in \ell^2$, the sequence y belongs to $D(H_\lambda)$. Symmetry gives the opposite inclusion of domains, and the actions agree there. For $S_N c = \sum_{n\leq N} c_n e_n$,

$$\|c - S_N c\|_{H_\lambda}^2 = \sum_{n>N} (1+\lambda_n^2)|c_n|^2 \longrightarrow 0.$$

□

For positive λ_n, the quadratic-form domain instead uses $\sum_n \lambda_n |c_n|^2 < \infty$. This is the domain of $H_\lambda^{1/2}$, not generally the domain of H_λ. With $\lambda_n = n$ and $c_n = n^{-s}$, the three thresholds are

membership	condition
ℓ^2	$s > 1/2$
$D(H_\lambda^{1/2})$	$s > 1$
$D(H_\lambda)$	$s > 3/2.$

Each equality case diverges harmonically. Thus a state with finite mean positive energy can still lack the Hilbert-space derivative required by the Schrödinger equation. The harmonic oscillator example has $\lambda_n = n + 1/2$ for $n \geq 0$ and the same asymptotic thresholds after replacing n by $n + 1$ in the coefficients.

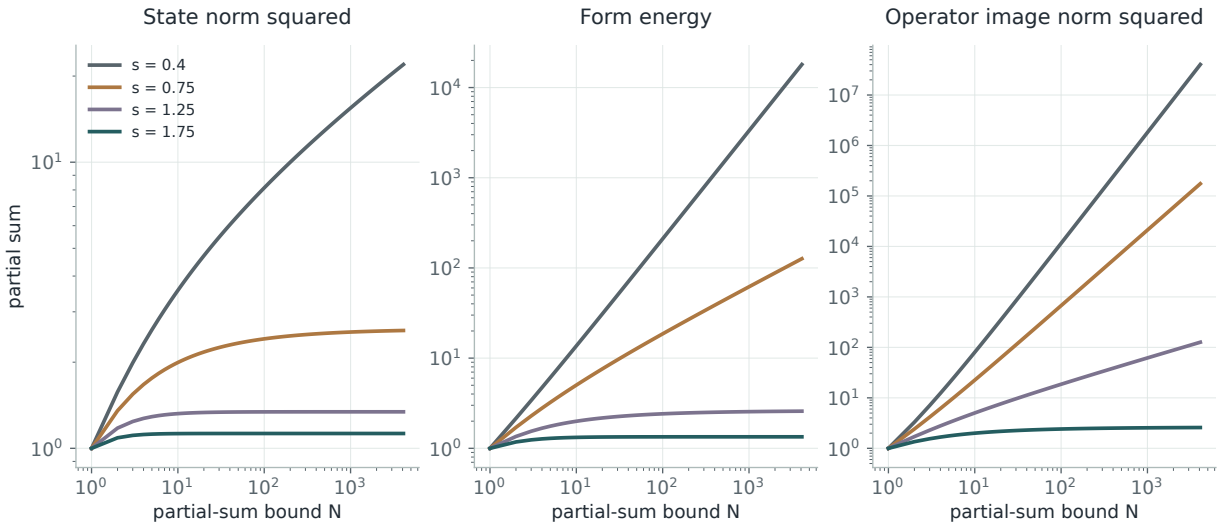


Figure 7. For $He_n = ne_n$, $n \geq 1$, coefficients $a_n = n^{-s}$ encounter three different thresholds: state norm, quadratic-form domain, and operator domain. The finite partial-sum curves illustrate the analytic p-series criteria.

12.2 Countable continuity and the stronger scalar-row argument

Suppose a supplied curve $\psi(t) \in D(H)$ is differentiable as a curve in $\mathcal{H}$ and satisfies $\dot{\psi} = -iH\psi$. For a domain-preserving atlas, put

$$p_b = \|P_b\psi\|^2, \qquad J_{ba} = 2\,\mathrm{Im}\,\langle P_b\psi, HP_a\psi\rangle\,.$$

Symmetry gives $J_{ba} = -J_{ab}$. Differentiating the bounded projected norm gives

$$\dot{p}_b = 2\,\mathrm{Im}\,\langle P_b\psi, H\psi\rangle\,.$$

Graph compatibility permits direct insertion of $HS_F\psi \to H\psi$. The formal development also proves a stronger statement for one receiving packet: symmetry moves H onto that packet, so only the strong state sum is needed. Indeed,

$$\langle P_b\psi, HP_a\psi\rangle = \langle HP_b\psi, P_a\psi\rangle\,,$$

and the sum over a converges to $\langle HP_b\psi, \psi\rangle = \langle P_b\psi, H\psi\rangle$. In fact the row is absolutely summable, since

$$\sum_a |\langle HP_b\psi, P_a\psi\rangle| = \sum_a |\langle P_aHP_b\psi, P_a\psi\rangle|$$
$$\leq \left(\sum_a \|P_aHP_b\psi\|^2\right)^{1/2} \left(\sum_a \|P_a\psi\|^2\right)^{1/2} = \|HP_b\psi\|\,\|\psi\|\,.$$

Consequently $\dot{p}_b = \sum_a J_{ba}$ without assuming graph compatibility. The two tail estimates describe the distinct arguments:

$$\left|\sum_{a\in F} J_{ba} - \dot{p}_b\right| \leq 2\,\|P_b\psi\|\,\|HS_F\psi - H\psi\|\,, \tag{37}$$

$$\left|\sum_{a\in F} J_{ba} - \dot{p}_b\right| \leq 2\,\|HP_b\psi\|\,\|S_F\psi - \psi\|\,. \tag{38}$$

The first bound uses graph-action control. The second uses symmetry and a fixed receiving packet. Neither states that the entire double array (J_{ba}) is absolutely summable.

12.3 Infinite coarse cells

For a partition $A = \bigsqcup_\alpha C_\alpha$, the strong sum $Q_\alpha = \sum_{a\in C_\alpha} P_a$ is a bounded orthogonal projection. Its weight is always

$$\|Q_\alpha\psi\|^2 = \sum_{a\in C_\alpha} \|P_a\psi\|^2\,.$$

To define a coarse current with $HQ_\alpha\psi$, additionally require $Q_\alpha D(H) \subset D(H)$. Fine domain preservation alone does not state this infinite-cell property.

Under this coarse-domain hypothesis, symmetry and the strong cell sums justify both iterated sums separately:

$$J^{\mathrm{c}}_{\beta\alpha} = 2\,\mathrm{Im}\,\langle Q_\beta\psi, HQ_\alpha\psi\rangle$$
$$= \sum_{b\in C_\beta}\left(\sum_{a\in C_\alpha} J_{ba}\right) = \sum_{a\in C_\alpha}\left(\sum_{b\in C_\beta} J_{ba}\right).$$

For example, first summing a moves H onto $P_b\psi$, then the outer sum pairs $\sum_b P_b\psi$ with the fixed vector $HQ_\alpha\psi$. The reverse order has its own symmetric argument. Flattening these expressions to a single unordered sum on $C_\beta \times C_\alpha$ requires joint summability. Equality of two justified iterated sums does not supply that hypothesis.

For a simple warning, the antisymmetric scalar array with $a_{2n,2n+1} = 1/n$, $a_{2n+1,2n} = -1/n$ for $n \geq 1$, and all other entries zero has finite support in each row but infinite total absolute sum. This array is only a summation example, not an asserted physical realization of the current. It shows why rowwise convergence alone cannot authorize arbitrary rearrangement.

The coarse balance is obtained by differentiating $\|Q_\beta\psi(t)\|^2$, using the bounded coarse projector. There is no need to interchange an infinite sum of derivatives. Likewise, the total packet norm is identified with $\|\psi(t)\|^2$ by Parseval before its conservation is established. The order of these arguments keeps the domain and summation obligations visible.

13 Constructing evolution and controlling truncation

A continuity theorem that assumes a differentiable solution does not construct one. For the real diagonal model, the construction is explicit and the precise differentiability domain can be proved. This supplies a complete small model in which the preceding distinctions can be inspected directly.

For real (λ_n) define

$$(U(t)c)_n = e^{-it\lambda_n} c_n, \qquad c \in \ell^2. \tag{39}$$

The modulus of every multiplier is one. Hence $U(t)$ is a unitary operator, $U(t+s) = U(t)U(s)$, and $U(-t) = U(t)^*$. For each fixed state it is strongly continuous. To prove the latter claim, split c into a finite head and a small ℓ^2 tail. The finite head is continuous coordinatewise, while the norm of the difference on the tail is at most twice its initial norm.

Proposition 13.1 (The exact differentiability domain)**.** *The orbit $t \mapsto U(t)c$ is differentiable in ℓ^2 at one, equivalently every, time precisely when $c \in D(H_\lambda)$. On that domain,*

$$\frac{d}{dt} U(t)c = -iH_\lambda U(t)c, \qquad H_\lambda U(t)c = U(t)H_\lambda c.$$

Proof. If $c \in D(H_\lambda)$, each coordinate difference quotient tends to $-i\lambda_n e^{-it\lambda_n} c_n$. The elementary bound $|e^{-iu} - 1| \le |u|$ bounds the quotient magnitude by $|\lambda_n c_n|$. The squared difference from its limit is bounded by $4\lambda_n^2 |c_n|^2$, a summable sequence. Dominated convergence therefore gives convergence in ℓ^2.

Conversely, if an ℓ^2 derivative d exists, bounded coordinate evaluation gives

$$d_n = -i\lambda_n e^{-it\lambda_n} c_n.$$

Since $d \in \ell^2$, one has $\sum_n \lambda_n^2 |c_n|^2 < \infty$. Commutation and domain preservation follow from the same coordinate formula. □

The constructed orbit is the unique global domain-valued solution with a given domain initial state. If two supplied solutions have the same initial value, their difference w satisfies $\dot{w} = -iH_\lambda w$. Symmetry gives

$$\frac{d}{dt} \|w\|^2 = 2 \operatorname{Re} \langle w, -iH_\lambda w \rangle = 0.$$

The initial zero norm remains zero. This uniqueness proof extends to supplied solutions of a symmetric operator, but the proof of existence above uses the explicit real diagonal construction.

For domain states, the orbit also satisfies the quantitative bound

$$\|U(t)c - U(s)c\| \le |t - s| \, \|H_\lambda c\| .$$

The constant belongs to the state. It is not an operator-norm bound for the unbounded generator. Spectral coordinate populations remain constant and their interpacket currents vanish. A different domain-preserving atlas can mix spectral coordinates and carry nonzero currents under the same evolution.

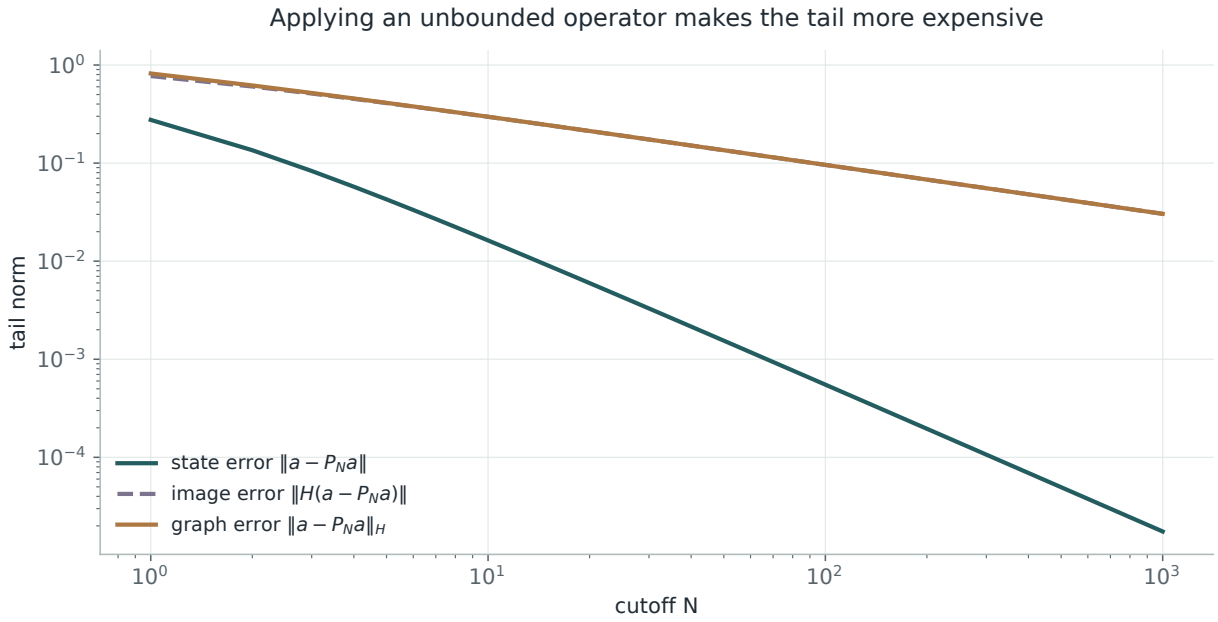


Figure 8. For the normalized state proportional to n^{-2} and $He_n = ne_n$, the state tail and generator tail decay at different rates. Graph control retains both contributions.

13.1 Uniformity in time for a fixed state

Coordinate cutoffs commute with $U(t)$. Thus for every $c \in \ell^2$,

$$\|U(t)c - S_N U(t)c\| = \|c - S_N c\|\,. \tag{40}$$

The right side tends to zero independently of $t \in \mathbb{R}$. This gives uniform convergence over all times for each fixed initial state. For $c \in D(H_\lambda)$, differentiating the finite cutoff orbit and the full orbit gives a second exact identity:

$$\left\| \frac{d}{dt} U(t)c - \frac{d}{dt} S_N U(t)c \right\| = \|H_\lambda c - S_N H_\lambda c\|\,. \tag{41}$$

This tail also tends to zero uniformly in time. State convergence alone does not imply this velocity convergence.

Example 13.2 (Strong convergence with no differentiable limiting orbit)**.** For $\lambda_n = n$, take $c_n = C/n$ with $C = (\sum_{n\geq 1} n^{-2})^{-1/2}$. This is a normalized ℓ^2 state outside $D(H_\lambda)$. Every finite cutoff orbit is differentiable, and (40) gives uniform convergence to the full orbit. Yet the full orbit is nowhere differentiable in ℓ^2, by the preceding proposition. In fact the norm of the cutoff velocity is $C\sqrt{N}$, which diverges.

For the faster normalized tail $c_n = C/n^2$, both errors are controlled:

$$\|c - S_N c\|^2 = C^2 \sum_{n>N} n^{-4}, \qquad \|H_\lambda c - S_N H_\lambda c\|^2 = C^2 \sum_{n>N} n^{-2}.$$

The state tail is $O(N^{-3/2})$ in norm, while the velocity tail is $O(N^{-1/2})$. A small state error can therefore coexist with a much larger derivative error even inside the true domain.

14 Which topology does reconstruction use?

Let S_N be orthogonal projections tending strongly to I. Their ranks need not be finite. For a bounded operator X, the reconstructed corner sum is $S_N X S_N$. Its natural convergence can be stronger when X belongs to a smaller operator class.

Theorem 14.1 (Three reconstruction statements)**.** *Under these hypotheses:*

1. *If X is bounded, $S_N X S_N \to X$ strongly.*
2. *If X is compact, $\|S_N X S_N - X\| \to 0$ in operator norm.*
3. *If X is Hilbert–Schmidt, $\|S_N X S_N - X\|_2 \to 0$ in Hilbert–Schmidt norm.*

Proof. Use the identity

$$S_N X S_N - X = S_N X (S_N - I) + (S_N - I) X.$$

Since $\|S_N\| \le 1$, application to a fixed x bounds the error by

$$\|X\| \, \|(S_N - I)x\| + \|(S_N - I)Xx\| \,,$$

which tends to zero.

For compact X, choose finite-rank Y with $\|X - Y\| < \varepsilon$. Strong convergence is uniform on each fixed finite-dimensional subspace, and self-adjointness of S_N controls the right error by the corresponding left error for Y^*. Thus $\|S_N Y S_N - Y\| \to 0$. The difference between the two reconstruction errors is at most 2ε, proving the claim.

For the last assertion, fix an orthonormal basis (e_j) and define

$$\|X\|_2^2 = \sum_j \|Xe_j\|^2 < \infty.$$

Each column $(S_N - I)Xe_j$ tends to zero, and its squared norm is bounded by $4\|Xe_j\|^2$. Dominated convergence gives $\|(S_N - I)X\|_2 \to 0$. The right error follows by taking adjoints, using $S_N^* = S_N$ and $\|X^*\|_2 = \|X\|_2$. The ideal inequality $\|AY\|_2 \le \|A\| \, \|Y\|_2$ and the displayed decomposition finish the proof. □

The Hilbert–Schmidt norm is independent of the chosen orthonormal basis, and it bounds the operator norm. It measures the total square size of all matrix columns. In a coordinate packet basis, it also measures the square sum of all matrix entries. Retaining a larger square of entries removes a correspondingly smaller summable tail. Operator norm instead measures the largest action on a unit vector, allowing the testing vector to depend on the cutoff.

Example 14.2 (The identity never has a small omitted tail in operator norm)**.** On an infinite-dimensional coordinate space, let S_N retain the first N coordinates. For $X = I$,

$$\|S_N I S_N - I\| = \|S_N - I\| = 1.$$

A unit vector e_{N+1} witnesses the lower bound. Nevertheless, $S_N x \to x$ for every fixed $x \in \ell^2$. This is the difference between pointwise control over states and uniform control over the full unit ball.

14.1 Zero padding and the correct asymptotic tail

For a diagonal bounded multiplier $f(H_\lambda)$, zero padding gives the exact formula

$$\|f(H_\lambda) - S_N f(H_\lambda) S_N\| = \sup_{n>N} |f(\lambda_n)|.$$

For the unitary multiplier $f(\lambda) = e^{-it\lambda}$, this norm is one at every finite cutoff. Equation (40) remains valid. Uniformity in time for a fixed state and failure of operator-norm convergence are therefore compatible.

For $\lambda_n = n$, a resolvent at $i\eta$, $\eta \neq 0$, has zero-padded error

$$\sup_{n>N} \frac{1}{\sqrt{n^2+\eta^2}} = \frac{1}{\sqrt{(N+1)^2+\eta^2}} \longrightarrow 0.$$

The heat multiplier e^{-tn} has error $e^{-t(N+1)}$ for fixed $t > 0$. These tails decay because the operators are compact. At $t = 0$ the heat operator is the identity, so the same norm-convergence assertion is false.

Two other bounded multipliers approach one along this positive spectrum:

$$b(n) = \frac{n}{\sqrt{1+n^2}}, \qquad q(n) = \frac{n-i}{n+i}.$$

Their correct anchored cutoff is

$$A_N(f) = S_N f(H_\lambda) S_N + (I - S_N).$$

The errors are then

$$\|b(H_\lambda) - A_N(b)\| = \sup_{n>N} \left| \frac{n}{\sqrt{1+n^2}} - 1 \right| \longrightarrow 0,$$

and

$$\|q(H_\lambda) - A_N(q)\| = \sup_{n>N} \frac{2}{\sqrt{n^2+1}} \longrightarrow 0.$$

The identity tail is an analytic choice justified by these asymptotic multipliers. It is not a universal prescription for every two-sided or non-diagonal spectrum.

On a graph-compatible atlas, an unbounded generator has another reconstruction statement:

$$S_N H S_N \psi - H\psi = S_N(HS_N\psi - H\psi) + (S_N - I)H\psi \longrightarrow 0, \qquad \psi \in D(H).$$

This is convergence of the displayed output vectors in the ambient Hilbert norm. It neither takes an operator norm of H nor asserts graph-norm convergence of those output vectors, which would require additional domain and second-action information.

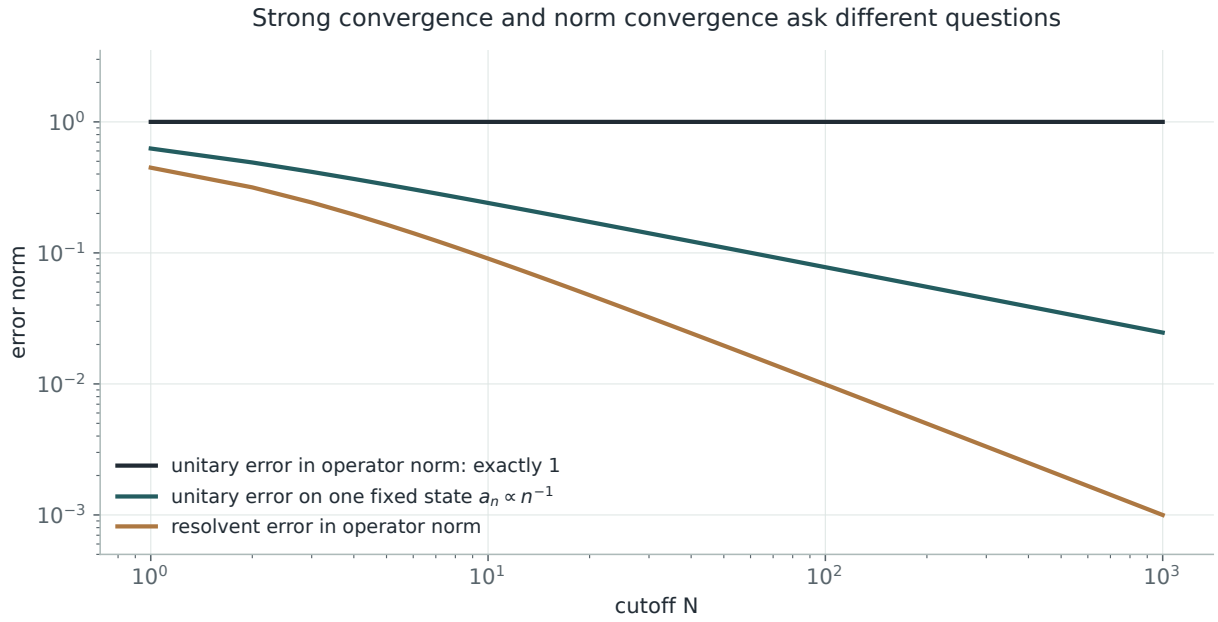


Figure 9. Zero-padded unitary cutoffs converge on a fixed state but retain operator-norm error one. For the same spectrum $n \geq 1$, the resolvent multiplier has a tail that decays in operator norm.

14.2 A bounded certificate for a nonreducing packet

A packet need not commute with an unbounded Hamiltonian. Suppose now that H is self-adjoint, P is a bounded orthogonal projection preserving $D(H)$, and $[H, P]$ on $D(H)$ extends to a bounded operator B. For nonreal z, let $R_z = (H - z)^{-1}$. It maps $\mathcal{H}$ into $D(H)$, so all products used below have their stated meaning. Multiplying the commutator identity on the domain by the resolvents gives

$$[R_z, P] = -R_z B R_z.$$

Indeed, for $x \in \mathcal{H}$, apply $H - z$ to $PR_z x$ to obtain $Px + BR_z x$, then apply R_z. Hence

$$\|[R_z, P]\| \leq \|R_z\|^2 \|B\|, \qquad \|[R_{i\eta}, P]\| \leq \frac{\|B\|}{\eta^2} \quad (\eta \neq 0).$$

The estimate converts a domain-sensitive commutator into a bounded one. In the generic formal certificate, the inverse and its domain properties are supplied explicitly. The diagonal modules also construct concrete inverses. Neither statement licenses a formal inverse for an unspecified operator.

15 The finite differential obstruction and matrix corners

Finite arrays support many useful difference operators. What they cannot support is a nonzero linear derivation of the full pointwise algebra $\mathbb{C}^N$ with its ordinary product. This obstruction explains why a finite operator calculus must declare which algebra carries its differentiation rule.

Theorem 15.1 (No nonzero derivation of the finite pointwise algebra)**.** *Let $D : \mathbb{C}^N \to \mathbb{C}^N$ be complex linear and satisfy*

$$D(fg) = D(f)g + fD(g)$$

for every pair of vectors, where multiplication is coordinatewise. Then $D = 0$.

Proof. For the coordinate idempotent e_a, the identity $e_a^2 = e_a$ gives $D(e_a) = 2e_aD(e_a)$. At a coordinate $b \neq a$, the right side is zero. At $b = a$, the equality reads $(D(e_a))_a = 2(D(e_a))_a$, again forcing zero. Thus $D(e_a) = 0$ for every basis vector, and linearity gives $D = 0$. □

The theorem concerns the semisimple pointwise algebra $\mathbb{C}^N$. It does not assert that every finite-dimensional commutative complex algebra has only the zero derivation. For example, on $\mathbb{C}[\varepsilon]/(\varepsilon^2)$, the map $D(a + b\varepsilon) = b\varepsilon$ is a nonzero derivation. The nilpotent element changes the algebraic hypothesis.

Example 15.2 (The precise defect of a forward difference)**.** On a cyclic grid, let $D_h f_j = (f_{j+1} - f_j)/h$, with $h \neq 0$. Expanding the product gives

$$D_h(fg)_j - (D_h f)_j g_j - f_j(D_h g)_j = h(D_h f)_j(D_h g)_j.$$

Thus the ordinary Leibniz identity fails unless the displayed product vanishes. A small defect on sufficiently regular sampled functions is compatible with the no-go theorem, because that theorem concerns an exact identity for every vector at fixed finite resolution.

15.1 Exact differentiation in the full matrix algebra

Move now to $M_N(\mathbb{C})$. To avoid confusing a Hermitian observable with the skew-adjoint state generator K, denote a Hermitian matrix by $B = B^*$ and define

$$\delta_B X = i[B, X] = i(BX - XB).$$

Then direct multiplication proves

$$\delta_B(XY) = (\delta_B X)Y + X\delta_B Y, \tag{42}$$

$$(\delta_B X)^* = \delta_B(X^*), \tag{43}$$

$$\operatorname{Tr}((\delta_B X)Y) = -\operatorname{Tr}(X\delta_B Y). \tag{44}$$

For the first identity, the middle terms $-iXBY$ and $iXBY$ cancel. For the second, taking the adjoint reverses the product order and conjugates i, with $B^* = B$ restoring the stated sign. For the third, cyclicity moves the leading B in $\operatorname{Tr}(BXY)$ to the end. The trace pairing is bilinear here. Combining (43) and (44) also gives the Hilbert–Schmidt identity

$$\langle \delta_B X, Y\rangle_{\mathrm{HS}} = -\langle X, \delta_B Y\rangle_{\mathrm{HS}}, \qquad \langle X, Y\rangle_{\mathrm{HS}} = \operatorname{Tr}(X^*Y).$$

These identities preserve multiplication order. They do not turn δ_B into a derivation from diagonal matrices to diagonal matrices. Usually it leaves that commutative subalgebra.

Example 15.3 (A nonzero derivative leaves the pointwise subalgebra)**.** For

$$B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

one obtains

$$\delta_B X = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \neq 0.$$

The output is off diagonal. This is exactly how the matrix algebra permits a nonzero derivation without violating the pointwise theorem.

For Hermitian generators B_i, B_j, put $\mathcal{F}_{ij} = i[B_i, B_j]$. This curvature matrix is Hermitian, and the Jacobi identity gives

$$[\delta_{B_i}, \delta_{B_j}]X = -[[B_i, B_j], X] = i[\mathcal{F}_{ij}, X] = \delta_{\mathcal{F}_{ij}} X. \tag{45}$$

Commuting generators produce commuting derivations. A nonzero curvature matrix need not act non-trivially on every observable: its action on a particular X vanishes precisely when that X commutes with it. This matrix identity provides an algebraic curvature mechanism. A physical geometric interpretation requires the additional structures developed in the gauge and gravity papers.

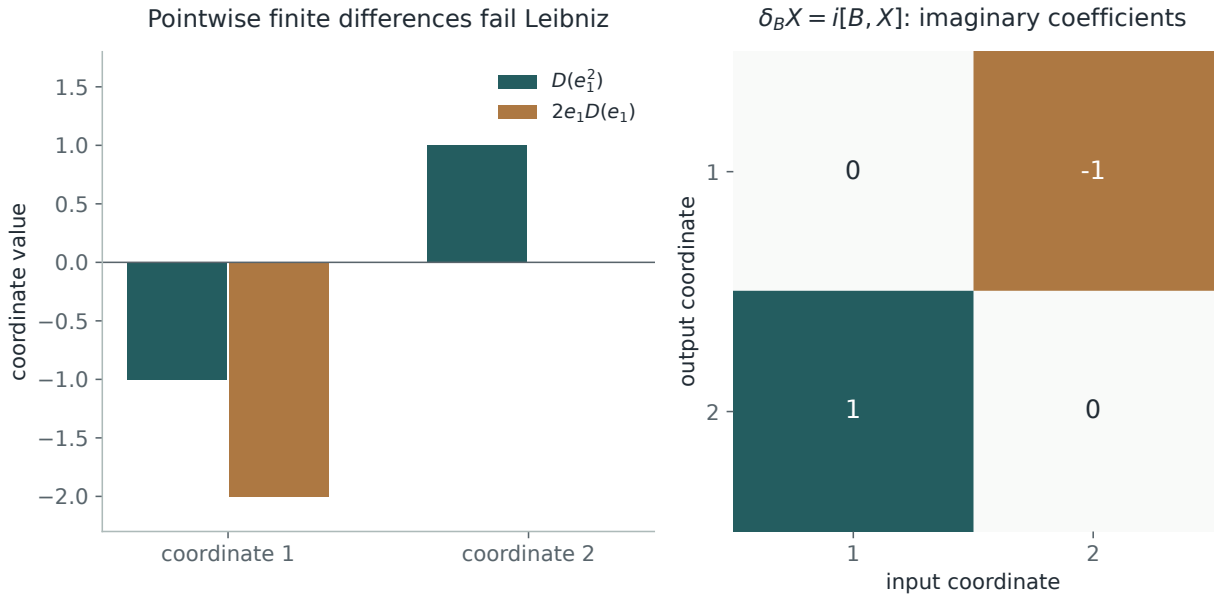


Figure 10. A finite pointwise difference has a Leibniz defect. Commutation inside a full matrix algebra obeys the Leibniz rule exactly, with its output allowed to leave the diagonal subalgebra.

15.2 Differentiated corner paths and coarse reconstruction

Let (P_a) be a finite sharp atlas. Writing $X_{ba} = P_b X P_a$ and $B_{ba} = P_b B P_a$, insertion of $I = \sum_c P_c$ gives

$$P_b \delta_B(X) P_a = i \sum_c \left(B_{bc} X_{ca} - X_{bc} B_{ca} \right). \tag{46}$$

Each term has a genuine intermediate packet c. One ordering applies X then B, while the other applies B then X. Replacing a matrix product by an entrywise product would destroy this path composition and the cancellation behind the Leibniz rule.

Summing the differentiated corners reconstructs $\delta_B X$. For a coarse partition with $Q_\alpha = \sum_{a \in C_\alpha} P_a$,

$$Q_\beta \delta_B(X) Q_\alpha = \sum_{\substack{b \in C_\beta \\ a \in C_\alpha}} P_b \delta_B(X) P_a.$$

This is exact aggregation of the already differentiated operator. It does not claim that discarding all internal matrix structure produces a new pointwise algebra with a nonzero exact derivation. The retained corners are the structure that makes the finite differential calculus work.

16 From packet calculus to field dynamics

The construction developed here has several exact outputs. It reconstructs operators from their transfer blocks. It converts skew-adjoint state dynamics into antisymmetric packet currents. It carries finite balance equations through partitions and moving frames. It identifies the additional information required by mixed states and open dynamics. It also gives precise versions of reconstruction beyond a finite atlas, where topology and domain membership become part of the statement.

The distinction between a supplied evolution and an existence theorem matters throughout the series. Differentiating a trajectory that already satisfies a state equation establishes the associated balance law. Constructing an evolution from a generator requires separate hypotheses and an existence argument. Identifying the generator of a previously constructed group is another specific task. Keeping these steps separate allows a later paper to reuse the balance identities without silently importing an unproved dynamics theorem.

The free Dirac monograph gives one concrete continuation. Fourier modes resolve the spinor field. Each mode carries a finite matrix, and a unitary Fourier map reconstructs the physical field. Finite packet consistency extends by Hilbert-space stability, while the maximal operator domain is recovered from weighted square summability. The Fourier construction provides the self-adjoint physical Hamiltonian. The following generator chapter proves its spectral-evolution identification at the level of mathematical analysis. Formal certification of that identification is recorded separately from the operator construction.

Gauge covariance and geometric refinement then extend the transport setting. Its task is to carry ordered transport through edge subdivision and cubical reconstruction while controlling the resulting defects. The moving-frame and transfer-block formulas in this volume supply part of its language. A regular prescribed connection, a coupled field evolution, and a constrained dynamical gauge theory each require their own precise mathematical setting.

The remaining parts of this book develop adic refinement, constrained gravity, and cosmological transport. Their use of the same resolution language does not eliminate their separate analytic and physical obligations. It makes those obligations comparable: specify the state space, identify the maps between resolutions, state the conserved or transported quantity, and exhibit the residual that prevents a stronger conclusion.

> **A reusable reading discipline**
>
> For each later result, locate the resolution, the transfer rule, the reconstruction statement, and the hypotheses that justify differentiation or passage to a limit. When a step fails, retain the explicit obstruction. It becomes part of the mathematical record and guides the next construction.

The same reading discipline applies across the book: exact algebra, analytic reconstruction, and physical interpretation must each carry their own hypotheses.

Sources and further reading

Project sources

[P1] Komorebi P. *Projector Refinement Physics, Volume I*. The foundational source studies cover transfer calculus, mixed and open dynamics, countable resolutions, unbounded domains, positive effects, and matrix-corner derivations. The technical source concordance identifies the relevant statements.

[P2] Komorebi P. *Projector Refinement Physics, Volume I: Technical Supplement*. Claim statements, formula registries, computational checks, and the hypotheses delimiting their conclusions.

[P3] Komorebi P. *Komorebi Physics formal source*. The foundations verification appendix gives declaration identities, retained compiler evidence, and the limits of its coverage.

[P4] Komorebi P. *Fourier Packets and the Continuum Dirac Operator*. The free-field construction in this book.

Classical context

[C1] G. Lindblad. "On the generators of quantum dynamical semigroups." *Communications in Mathematical Physics* **48** (1976), 119–130. https://doi.org/10.1007/BF01608499.

[C2] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan. "Completely positive dynamical semigroups of N-level systems." *Journal of Mathematical Physics* **17**, no.5 (1976), 821–825. https://doi.org/10.1063/1.522979.

[C3] John Watrous. *Theory of Quantum Information*, CS766/QIC820, Fall2011, Lecture5, Section5.1, Theorem5.1. The finite-dimensional Naimark construction is given by the square-root isometry and coordinate projectors. https://cs.uwaterloo.ca/~watrous/TQI-notes/TQI-notes.05.pdf.

The first two classical references identify the standard setting for the finite Lindblad expression used in this volume. Their general classification results are not premises of the elementary packet calculations proved here. The Watrous notes give a direct comparison for the finite dilation construction. The source-check record distinguishes verified bibliographic metadata, the inspected dilation proof, and the inaccessible full AIP article.

This bibliography provides context and traceability. It is not an exhaustive literature review or a claim of historical priority. The mathematical review concerns the displayed arguments and the explicitly recorded source correspondence.